SOFTWARE ENGINEERING 2: PROBLEM PROJECT PROPOSAL

Research Group Dissertation & Presentation

Client: Amenda Raganya

Group Allocation: Group 3 – Cape Town Campus

Group Members: Willem Basson, Dylan Shepard, Moegammad Green, Sihle Kayise

Submission Data: 2 October 2026

Selected Group Lead

- **Group Lead:** Willem Basson
- **Coordination Role:** Responsible for client communication via Microsoft Teams, milestone tracking, repository management, and consolidating dissertation deliverables across the four members.

Problem Statement

- **Domain:** Higher Education Administration & Workplace Compliance
- **Background & Context:** Occupational and technical qualifications require students to complete mandatory practical workplace hours, workshops, and portfolio submissions before qualifying for final external summative assessments. Currently, tracking these milestones relies heavily on fragmented manual processes—including static PDF forms, emailed spreadsheets, and physical sign-offs between students, academic supervisors, and host employers.
- **Core Problem:** This decentralized workflow creates critical bottlenecks: lost or unverified logbook entries, delayed identification of at-risk students, lack of real-time visibility for academic coordinators, and high administrative overhead during audit cycles. Existing Learning Management Systems (LMS) focus on classroom grading and lack dedicated multi-party workflows for external employer verification.
- **Proposed Software Engineering Solution:** The group proposes architecting and evaluating **TrackWIL**, a centralized, role-based web application that digitizes workplace logbook submissions, automates supervisor verification via secure token links (requiring no complex onboarding for external employers), and provides campus coordinators with a real-time compliance analytics dashboard.
- **Research & Data Feasibility:** Empirical data and system requirements will be gathered directly from campus academic staff, placement coordinators, and enrolled students via structured questionnaires, time-on-task comparisons (manual vs. digital submission), and usability testing.

Target Stakeholders

- **Primary Users (Students):** Require a mobile-responsive interface to log daily/weekly practical tasks, upload evidence documents, and track their completion percentage against qualification requirements.
- **External Evaluators (Host Employers / Mentors):** Require a frictionless, secure verification workflow to review and digitally sign off on student hours without needing to manage permanent institutional accounts.
- **Institutional Administrators (Academic Coordinators & Assessors):** Require automated exception reporting (flagging overdue submissions), bulk compliance exports, and tamper-evident audit logs for external quality assurance bodies.

High-Level Software Engineering Methodology

To balance structured academic research documentation with iterative software architecture, Group 3 will adopt a **Hybrid Agile-Scrum Methodology** organized around a 4+1 Architectural View Model:

- **Requirements Engineering & Domain Modeling (Weeks 1–2):** Conduct stakeholder interviews and survey data collection to establish functional and non-functional requirements, documented via UML Use Case and Activity diagrams.
- **Iterative System Architecture & Prototyping (Weeks 3–5):** Execute two-week Scrum sprints utilizing GitHub for version control and project board tracking. The system will be designed using a 3-tier architecture (Relational Database with strict referential integrity, RESTful API backend with Role-Based Access Control, and a responsive web frontend).
- **Empirical Evaluation & Verification (Weeks 6–7):** Perform unit and integration testing alongside System Usability Scale (SUS) testing with target stakeholders to empirically evaluate efficiency gains against existing manual workflows for the final dissertation and presentation.